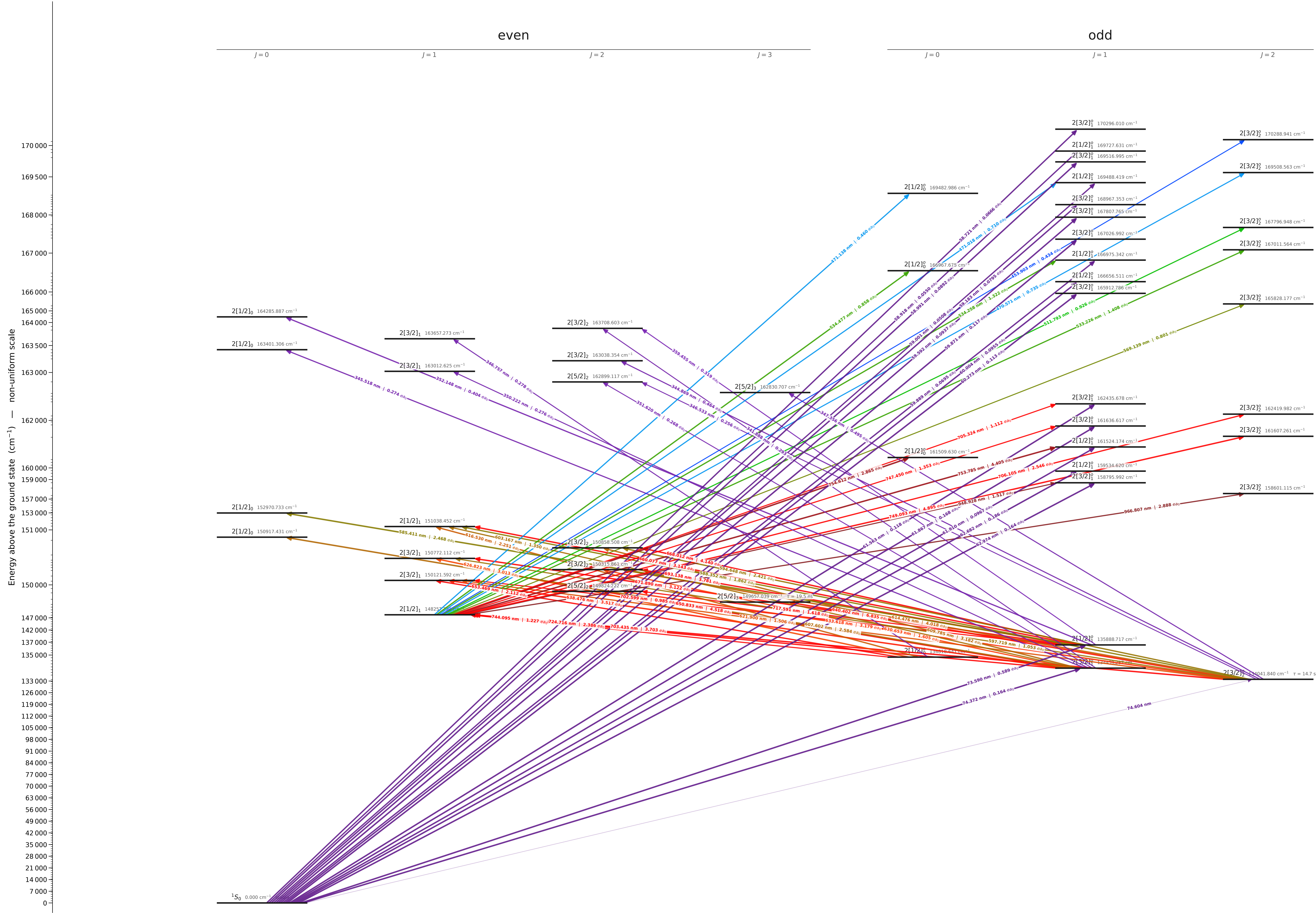
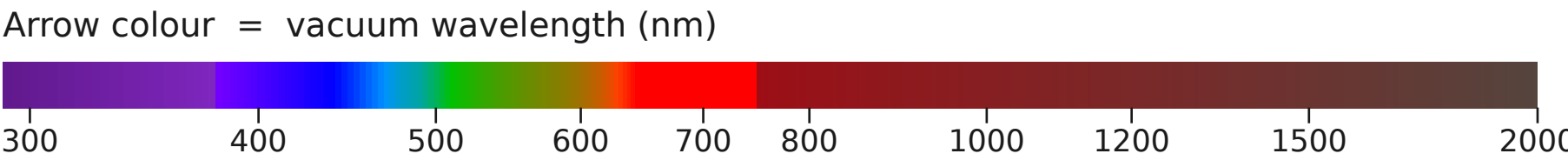


Neon

Ne (Ne I). Strongest classified transitions below 2 μm (NIST): 49 levels, 71 lines.

Ne⁺ ionisation limit 173929.75 cm⁻¹ = 21.56454 eV (57.494 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U||e r||J\rangle$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ U e r J\rangle$ ($e a_0$)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
¹ S ₀ – 2[3/2] ₂ ^o	74.6036	–	4018.4733	–	6.790e-02	1.08e-08	100	74.6 nm M2 decay of the metastable 3P2 state
2[3/2] ₂ ^o – 2[5/2] ₃	640.4017	640.2247	468.1319	6.835	5.149e+07	8.162	100	640.2 nm laser-cooling / MOT line

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

²¹Ne hyperfine structure ($I = 3/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
2[3/2] ₂ ^o	-267.68(3)	-111.55(10)	F=1/2: +1106.95 F=3/2: +803.04 F=5/2: +203.56 F=7/2: -830.93
2[5/2] ₃	-142.40(20)	-107.7(11)	F=3/2: +789.78 F=5/2: +514.55 F=7/2: +53.85 F=9/2: -667.73

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
2[3/2] ₂ ^o – 2[5/2] ₃	640.402	22-20	1625.90(15)
2[3/2] ₂ ^o – 2[5/2] ₃	640.402	21-20	855.7(10)
2[3/2] ₂ ^o – 2[5/2] ₃	640.402	22-21	770.3(10)

Shift = $\nu(\text{first isotope}) - \nu(\text{second isotope})$.